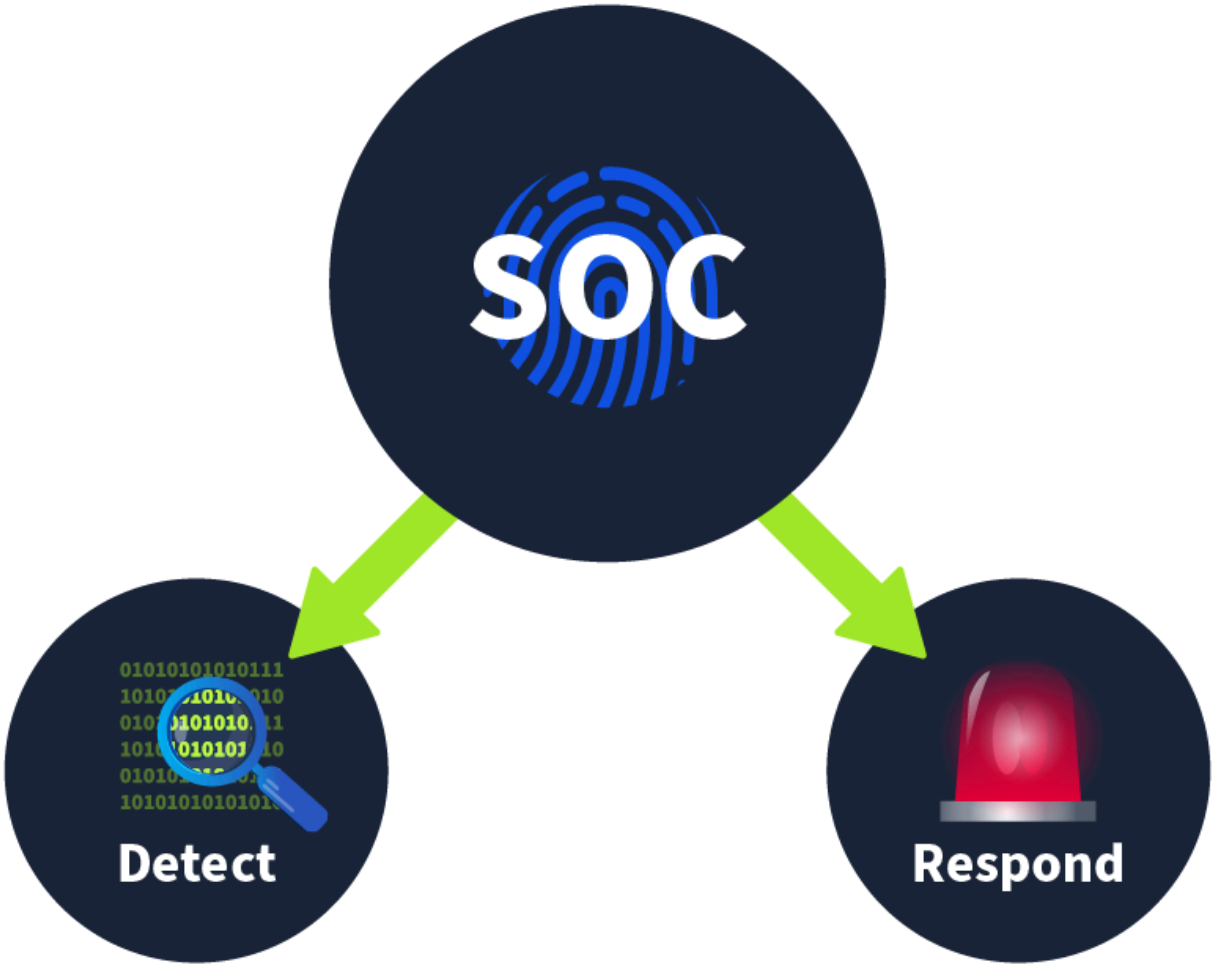
SOC Fundamentals

A SOC (Security Operations Center) is a dedicated facility operated by a specialized security team. This team aims to continuously monitor an organization's network and resources and identify suspicious activity to prevent damage. This team works 24 hours a day, seven days a week.

Learning Objectives

- Building a baseline for SOC (Security Operations Center)
- Detection and response in SOC
- The role of People, Processes, and Technology
- Practical exercise

The main focus of the SOC team is to keep **Detection** and **Response** intact



Detection

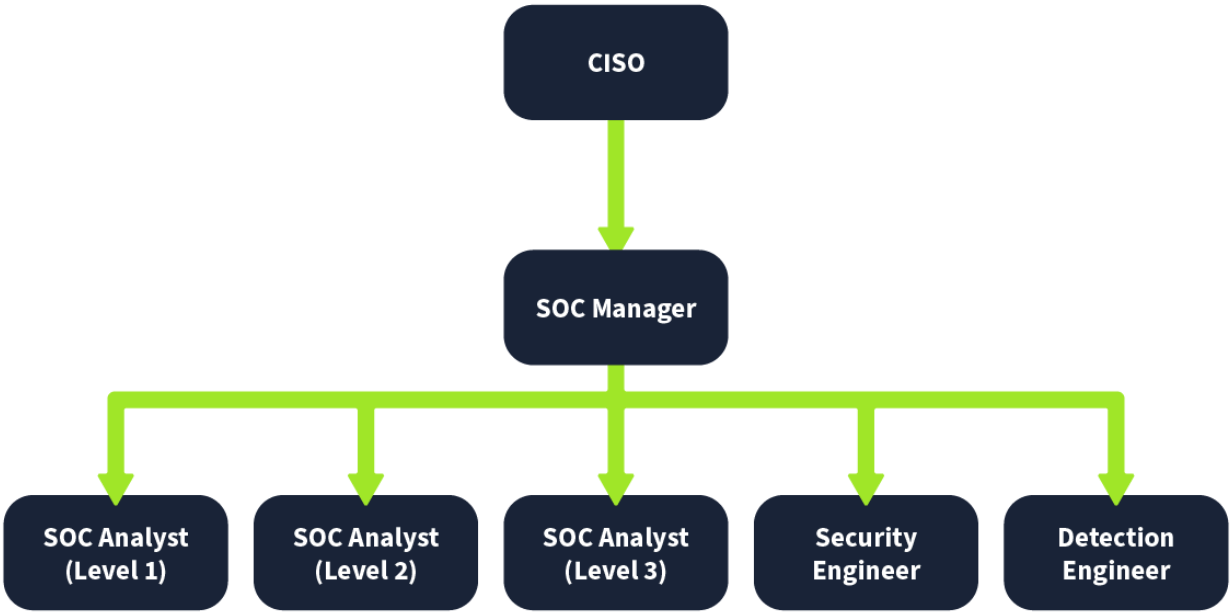
- **Detect vulnerabilities**: A vulnerability is a weakness that an attacker can exploit to carry out things beyond their permission level. A vulnerability might be discovered in any device's software (operating system and programs), such as a server or computer. For instance, the SOC might discover a set of MS Windows computers that must be patched against a specific published vulnerability. Strictly speaking, vulnerabilities are not necessarily the SOC's responsibility; however, unfixed vulnerabilities affect the security level of the entire company.
- **Detect unauthorized activity**: Consider the case where an attacker discovered the username and password of one of the employees and used them to log in to the company system. It is crucial to detect this kind of unauthorized activity quickly before it causes any damage. Many clues, such as geographic location, can help us detect this.
- **Detect policy violations**: A security policy is a set of rules and procedures created to help protect a company against security threats and ensure compliance. What is considered a violation would vary from company to company; examples include downloading pirated media files and sending confidential company files insecurely.
- **Detect intrusions**: Intrusions refer to unauthorized access to systems and networks. One scenario would be an attacker successfully exploiting our web application. Another would be a user visiting a malicious site and getting their computer infected.

Response

- **Support with the incident response**: Once an incident is detected, certain steps are taken to respond to it. This response includes minimizing its impact and performing the root cause analysis of the incident. The SOC team also helps the incident response team carry out these steps.

There are three pillars of a SOC. With all these pillars, a SOC team becomes mature and efficiently detects and responds to different incidents. These pillars are **People**, **Process**, and **Technology**.

The **People** are known as the SOC team. This team has the following roles and responsibilities.



- SOC **Analyst (Level 1):** Anything detected by the security solution would pass through these analysts first. These are the first responders to any detection. SOC Level 1 Analysts perform basic alert triage to determine if a specific detection is harmful. They also report these detections through proper channels.
- SOC **Analyst (Level 2):** While Level 1 does the first-level analysis, some detections may require deeper investigation. Level 2 Analysts help them dive deeper into the investigations and correlate the data from multiple data sources to perform a proper analysis.
- SOC **Analyst (Level 3):** Level 3 Analysts are experienced professionals who proactively look for any threat indicators and support in the incident response activities. The critical severity detection reported by Level 1 and Level 2 Analysts are often security incidents that need detailed responses, including containment, eradication, and recovery. This is where Level 3 analysts' experience comes in handy.
- **Security Engineer:** All analysts work on security solutions. These solutions need deployment and configuration. Security Engineers deploy and configure these security solutions to ensure their smooth operation.
- **Detection Engineer:** Security rules are the logic built behind security solutions to detect harmful activities. Level 2 and 3 Analysts often create these rules, while the SOC team can sometimes also utilize the detection engineer role independently for this responsibility.

- **SOC Manager:** The SOC Manager manages the processes the SOC team follows and provides support. The SOC Manager also remains in contact with the organization's CISO (Chief Information Security Officer) to provide him with updates on the SOC team's current security posture and efforts.

Alert Triage

The alert triage is the basis of the SOC team. The first response to any alert is to perform the triage. The triage is focused on analyzing the specific alert. This determines the severity of the alert and helps us prioritize it. The alert triage is all about answering th

Alert: Malware detected on Host: GEORGE PC

5 Ws	Answers
What?	A malicious file was detected on one of the hosts inside the organization's network.
When?	The file was detected at 13:20 on June 5, 2024.
Where?	The file was detected in the directory of the host: "GEORGE PC".
Who?	The file was detected for the user George.
Why?	After the investigation, it was found that the file was downloaded from a pirated software-selling website. The investigation with the user revealed that they downloaded the file as they wanted to use a software for free.

Reporting

The detected harmful alerts need to be escalated to higher-level analysts for a timely response and resolution. These alerts are escalated as tickets and assigned to the relevant people. The report should discuss all the 5 Ws

along with a thorough analysis, and screenshots should be used as evidence of the activity.

Incident Response and Forensics

Sometimes, the reported detections point to highly malicious activities that are critical. In these scenarios, high-level teams initiate an incident response. The incident response process is discussed in detail in the Incident Response room. A few times, a detailed forensics activity also needs to be performed. This forensic activity aims to determine the incident's root cause by analyzing the artifacts from a system or network.

The **Technology** portion in the SOC pillars refers to the security solutions. These security solutions efficiently minimize the SOC team's manual effort to detect and respond to threats.

An organization's network consists of many devices and applications. As a security team, individually detecting and responding to threats in each device or application would require significant effort and resources. Security solutions centralize all the information of the devices or applications present in the network and automate the detection and response capabilities.

Let's get a brief understanding of some of these security solutions:

- **SIEM:** Security Information and Event Management (SIEM) is a popular tool used in almost every SOC environment. This tool collects logs from various network devices, referred to as log sources. Detection rules are configured in the SIEM solution, which contains logic to identify suspicious activity. The SIEM solution provides us with the detections after correlating them with multiple log sources and alerts us in case of a match with any of the rules. Modern SIEM solutions surpass this rule based detection analysis, providing us with user behavior analytics and threat intelligence capability. Machine learning algorithms support this to enhance the detection capabilities.

Note: The SIEM solution only provides the **Detection** capabilities in a SOC environment.

- **EDR:** Endpoint Detection and Response (EDR) provides the SOC team with detailed real-time and historical visibility of the devices' activities. It operates on the endpoint level and can carry out automated responses. EDR has extensive detection capabilities for endpoints, allowing you to investigate them in detail and respond with a few clicks.
- **Firewall:** A firewall functions purely for network security and acts as a barrier between your internal and external networks (such as the Internet). It monitors incoming and outgoing network traffic and filters any unauthorized traffic. The firewall also has some detection rules deployed, which help us identify and block suspicious traffic before it reaches the internal network.

Several other security solutions play unique roles in a SOC environment, such as Antivirus, EPP, IDS/IPS, XDR, SOAR, and more. The decision on what Technology to deploy in the SOC comes after careful consideration of the threat surface and the available resources in the organization.